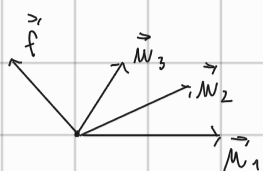


RAPPRESENTAZIONE MATRICIALE DI OPERATORI e FUNZIONI D'ONDA:

• Spaz. Vett. $\mathbb{C}^3$



$$\vec{m}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\vec{m}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\vec{m}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

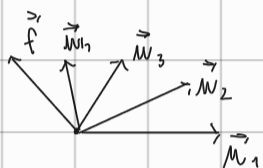
$$\vec{f} = \begin{pmatrix} f_x \\ f_y \\ f_z \end{pmatrix} = |\vec{f}\rangle \Leftrightarrow \langle \vec{f} | = |\vec{f}\rangle^{*T}$$

$$\langle \vec{g} | \vec{f} \rangle = \langle \vec{f} | \vec{g} \rangle^*$$

$$|\vec{m}_n\rangle \Rightarrow |\vec{f}\rangle = \sum_n c_n |\vec{m}_n\rangle, \quad \langle \vec{m}_n | \vec{m}_m \rangle = \delta_{nm}$$

MATRICI $\hat{M}$

Spazi Funz.: $\mathbb{L}^2$



$$|f\rangle \Leftrightarrow \langle f |^*$$

$$\langle f | g \rangle = \int f^* g dx$$

$$\{m_n\} \Leftrightarrow |f\rangle = \sum_n c_n |m_n\rangle$$

Stessa struttura

OPERATORI $\hat{O}$

(unisco i due)

=> SPAZIO ASTRATTO dei KET H

↳ Spazio vett. complessa $|f\rangle$

$$|\chi\rangle \in |f\rangle \Rightarrow |\chi\rangle = \alpha |\psi\rangle + \beta |f\rangle \quad \text{LINEARITÀ} \quad \alpha, \beta \in \mathbb{C}$$

$$\text{Ad ogni } |\cdot\rangle \Leftrightarrow \langle \cdot | = |\cdot\rangle^{*T} = |\cdot\rangle^\dagger$$

• Prodotto interno $\Leftrightarrow$ Prodotto scalare

$$\langle f | \psi \rangle = \langle \psi | f \rangle^*$$

• Lo spaz. dei KET si dice completo rispetto all'a norma indotta dal prodotto scalare.

$$\|\psi\| = \sqrt{\langle \psi | \psi \rangle}$$

$$\cdot \left\{ |n\rangle \right\}^{\text{BASE}} \text{ t.c. } |\psi\rangle = \sum_n c_n |n\rangle$$

(finalmente non scelgo una base precisa
: ket rimangono ogg. astratti)

SCELGO UNA BASE: $|m_i\rangle$

$$|f\rangle = \sum_i c_i |m_i\rangle \quad c_i = \langle m_i | f \rangle$$

$$|\vec{v}\rangle = 2|\vec{m}_x\rangle + 3|\vec{m}_y\rangle + 4|\vec{m}_z\rangle \Rightarrow |\vec{v}\rangle = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}$$

$$|\psi\rangle = \sqrt{\frac{3}{5}}|Y_1^1\rangle + \sqrt{\frac{1}{5}}|Y_1^0\rangle + \sqrt{\frac{1}{5}}|Y_1^{-1}\rangle \quad \begin{cases} l=1 \\ m_l=1 \end{cases}$$

$$|\psi\rangle = \begin{pmatrix} \sqrt{3/5} \\ \sqrt{1/5} \\ \sqrt{1/5} \end{pmatrix} \quad \begin{array}{l} \text{1a vett. dei coef. SOSTITUIRE} \\ \text{una BASE dettata. univ. vec.} \\ \text{una BASE} \end{array}$$

$$\Rightarrow |\psi\rangle \rightarrow \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \\ c_n \end{pmatrix} \quad \text{n comp. della base}$$

$$|\varphi\rangle = |\hat{O}\psi\rangle = \{ |w_n\rangle \} = \hat{O} \sum_j c_j |w_j\rangle = \sum_j c_j \hat{O} |w_j\rangle$$

$$\Rightarrow \langle w_i | \varphi \rangle = \langle w_i | \hat{O} \psi \rangle = \langle w_i | \sum_j c_j \hat{O} |w_j\rangle \Rightarrow b_i = \langle w_i | \varphi \rangle = \sum_j c_j \langle w_i | \hat{O} |w_j\rangle$$

Proiezione di $|\varphi\rangle$ sul ket di base w_i

$$b_i = (\phi)_i = \sum_j c_j \langle w_i | \hat{O} |w_j\rangle$$

$$O_{ij} = \langle w_i | \hat{O} |w_j\rangle$$

$$\Rightarrow b_i = (\phi)_i = \sum_j O_{ij} c_j$$

$$\begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_i \\ \vdots \end{pmatrix} = \begin{pmatrix} (\hat{O}\psi)_1 \\ \vdots \\ (\hat{O}\psi)_i \\ \vdots \end{pmatrix} = \begin{pmatrix} O_{11} & O_{12} & \dots & O_{1j} & \dots \\ \vdots & \vdots & & \vdots & \dots \\ O_{i1} & & & O_{ij} & \dots \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_j \\ \vdots \end{pmatrix}$$

↓
Vett. proiezione $|\varphi\rangle$
su $\langle w$

$$b_1 = \sum_j O_{1j} c_j = O_{11}c_1 + O_{12}c_2 + \dots$$

$|\varphi\rangle = |\hat{O}\psi\rangle \Leftrightarrow \varphi = \hat{O}\psi \Rightarrow$ Ho trasf. questo operaz. diff. in una matrice per un vettore

Rappr. di $|\psi\rangle$
sulla base $|w_i\rangle$

MATRICE di $\hat{O}$ sulla
base $|w_i\rangle$

Rappresent. di $|\psi\rangle$ sulla
base $|w_i\rangle$

$$\begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_i \\ \vdots \end{pmatrix} = \begin{pmatrix} (\hat{O}\psi)_1 \\ \vdots \\ (\hat{O}\psi)_i \\ \vdots \end{pmatrix} = \begin{pmatrix} O_{11} & O_{12} & \dots & O_{1j} & \dots \\ \vdots & \vdots & & \vdots & \ddots \\ \vdots & & & O_{ij} & \dots \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_i \\ \vdots \end{pmatrix}$$

BASE N $|w_i\rangle \Rightarrow \hat{O} \ N \times N$

$$|\psi\rangle = \sum_i c_i |w_i\rangle = \sum_i |w_i\rangle \langle w_i | \psi \rangle = \sum_i \underbrace{(|w_i\rangle \langle w_i|)}_{\mathbb{I} \text{ (op. identit.)}} |\psi\rangle$$

$\langle \hat{w} | \hat{O} | w_j \rangle \Rightarrow$ Elemento di matr. di $\hat{O}$

$\mathbb{I}$ (op. identit.)

$$\langle w_i | \hat{O} \hat{P} | w_j \rangle = \sum_k \langle w_i | \hat{O} | w_k \rangle \langle w_k | \hat{P} | w_j \rangle = \sum_k O_{ik} P_{kj}$$

$$(\hat{O} \hat{P})_{ij} = \sum_k O_{ik} P_{kj}$$

$$\begin{pmatrix} (OP)_{11} & (OP)_{12} & \dots \\ \vdots & \vdots & \\ (OP)_{i1} & (OP)_{ij} & \dots \end{pmatrix} = \begin{pmatrix} O_{11} & \dots & O_{1j} & \dots \\ \vdots & & \vdots & \\ O_{i1} & \dots & O_{ij} & \dots \end{pmatrix} \begin{pmatrix} P_{11} & \dots & P_{1j} & \dots \\ \vdots & & \vdots & \\ P_{i1} & \dots & P_{ij} & \dots \end{pmatrix}$$

$$(OP)_{11} = \sum_k O_{1k} P_{k1} = O_{11} P_{11} + O_{12} P_{21} + O_{13} P_{31} + \dots$$

Il prod. tra due op.
è diventato un prod. tra
matrici

$$(\hat{O}^+)_{ji} = \langle w_i | \hat{O}^+ | w_j \rangle = \langle \hat{O} w_i | w_j \rangle = \langle w_j | \hat{O} | w_i \rangle^* = O_{ji}^*$$

1) $\hat{O} \psi \Rightarrow [\hat{O}] [\psi]$ ^{matrice}

2) $\hat{O} \hat{P} \Rightarrow [\hat{O}] [\hat{P}]$

3) $\hat{O}^+ \Rightarrow O_{ji}^*$

$$(1) |m_j\rangle = \text{SET di AUTOSTATI di } \hat{O} \Rightarrow \langle m_i | \hat{O} | m_j \rangle = 0_j \delta_{ij}$$

$$(2) \hat{O} \Rightarrow \text{SPETTRO di AUTOSTATI DISCRETO} \Rightarrow \hat{O} \Rightarrow \text{MAT } N \times N$$

$$\hat{O} \Rightarrow \text{SPETTRO CONTINUO} \Rightarrow \text{MAT. } \times \text{ di } \hat{O} \text{ (inf. non numerabile)}$$

$$\hat{H} \Rightarrow \text{SCELGO una base qualunque} \Rightarrow \langle m_i | \hat{H} | m_j \rangle \begin{pmatrix} H_{11} & H_{12} & \dots \\ \vdots & & \end{pmatrix}$$

sono mat. che mi interessano $\Rightarrow$ blocchi

(mi interessano solo alcuni valori)

Es. di rappresent. matriciale:

$$\Rightarrow \text{Matrici di } \hat{L}_z \text{ per } l=1 \quad Y_1^1 \Rightarrow |1,1\rangle, Y_1^0 \Rightarrow |1,0\rangle, Y_1^{-1} \Rightarrow |1,-1\rangle$$

$$|\psi\rangle = \sum_{l,m} c_{l,m} |l,m\rangle \Rightarrow |\psi\rangle = \sum_m c_m |\bar{l}, m\rangle$$

$$|\psi\rangle = \alpha |1,1\rangle + \beta |1,0\rangle + \gamma |1,-1\rangle \Rightarrow \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}$$

$$\hat{L}_z |l,m\rangle = \hbar m |l,m\rangle \Rightarrow \hat{L}_z |1,m\rangle = \hbar m |1,m\rangle$$

$$\langle 1, m' | \hat{L}_z |1,m\rangle = \hbar m \delta_{m,m'} \Leftrightarrow \langle 1, m' | \hbar m |1,m\rangle$$

$$\hat{L}_z = -i\hbar \frac{\partial}{\partial \phi} \Rightarrow L_z = \begin{pmatrix} \langle 1,1 | \hat{L}_z |1,1\rangle & \langle 1,1 | \hat{L}_z |1,0\rangle & \langle 1,1 | \hat{L}_z |1,-1\rangle \\ \langle 1,0 | \hat{L}_z |1,1\rangle & \langle 1,0 | \hat{L}_z |1,0\rangle & \langle 1,0 | \hat{L}_z |1,-1\rangle \\ \langle 1,-1 | \hat{L}_z |1,1\rangle & \langle 1,-1 | \hat{L}_z |1,0\rangle & \langle 1,-1 | \hat{L}_z |1,-1\rangle \end{pmatrix} =$$

$$= \begin{pmatrix} \hbar & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\hbar \end{pmatrix} = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}$$

Trovo $\hat{L}_x$ e $\hat{L}_y$

$$\hat{L}_+ |l,m\rangle = \hbar \sqrt{l(l+1) - m(m+1)} |l, m+1\rangle$$

$$\hat{L}_+^\dagger = \hat{L}_-$$

$$\hat{L}_- |l,m\rangle = \hbar \sqrt{l(l+1) - m(m-1)} |l, m-1\rangle$$

$$\bullet \langle l, m' | \hat{L}_+ | l, m \rangle = \sqrt{l(l+1) - m(m+1)} \hbar \delta_{m', m+1}$$

$$\bullet \langle l, m' | \hat{L}_- | l, m \rangle = \sqrt{l(l+1) - m(m-1)} \hbar \delta_{m', m-1}$$

$$\hat{L}_+ = \begin{pmatrix} \langle 1, 1 | \hat{L}_+ | 1, 1 \rangle & \langle 1, 1 | \hat{L}_+ | 1, 0 \rangle & \langle 1, 1 | \hat{L}_+ | 1, -1 \rangle \\ \langle 1, 0 | \hat{L}_+ | 1, 1 \rangle & \langle 1, 0 | \hat{L}_+ | 1, 0 \rangle & \langle 1, 0 | \hat{L}_+ | 1, -1 \rangle \\ \langle 1, -1 | \hat{L}_+ | 1, 1 \rangle & \langle 1, -1 | \hat{L}_+ | 1, 0 \rangle & \langle 1, -1 | \hat{L}_+ | 1, -1 \rangle \end{pmatrix} = \begin{pmatrix} 0 & \sqrt{2} & 0 \\ 0 & 0 & \sqrt{2} \\ 0 & 0 & 0 \end{pmatrix} \hbar$$

$$\hat{L}_- = \hbar \begin{pmatrix} 0 & 0 & 0 \\ \sqrt{2} & 0 & 0 \\ 0 & \sqrt{2} & 0 \end{pmatrix}$$

$$\hat{L}_x = \frac{1}{2} (\hat{L}_+ + \hat{L}_-) = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\hat{L}_y = \frac{1}{2i} (\hat{L}_+ - \hat{L}_-) = \frac{\hbar}{2i} \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$\Gamma l=2 \Rightarrow m_l = -2, -1, 0, 1, 2$$

$$S = \frac{1}{2} \quad m_s = \pm \frac{1}{2} \quad |\uparrow\rangle, |\downarrow\rangle \quad |\chi\rangle = c_1 |\uparrow\rangle + c_2 |\downarrow\rangle \Rightarrow |\chi\rangle = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

OPERAT. A SPECTRO CONT.:

$$|\psi\rangle = \sum_i c_i |m_i\rangle \quad |m_i\rangle \Rightarrow |x\rangle$$

$$\downarrow$$

$$|\psi\rangle = \int c(x) |x\rangle dx \quad , \quad \langle x | \psi \rangle = \int dx' c(x') \underbrace{\langle x | x' \rangle}_{\delta(x-x')}$$

$$\langle x | \psi \rangle = \int dx' c(x') \delta(x-x') = c(x)$$

$|c(x)|^2 \Rightarrow$ Dens. probab. corrispond. al ket $|x\rangle$

$$\Rightarrow |c(x)|^2 = |\psi(x)|^2 \hookrightarrow c(x) = \psi(x) \Rightarrow \boxed{\psi(x) = \langle x | \psi \rangle}$$

$$|\psi\rangle \Rightarrow \begin{pmatrix} \psi(x_1) \\ \psi(x_2) \\ \vdots \\ \psi(x_n) \end{pmatrix} \quad \text{infin. non numerabile}$$

TEORIA DELLE PERTURB. NON DIPENDENTI DAL TEMPO: STATI NON DEGENERI

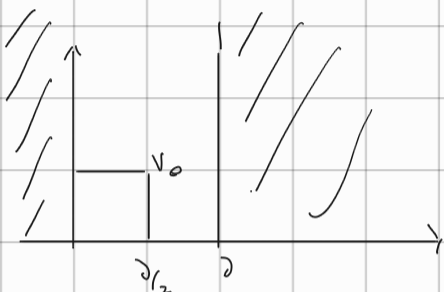
$\hat{H}^{(0)}$ HAMILTON. IMPERTURBATA } LO RISORNO ESATTO. $\Rightarrow \hat{H}^{(0)} w_n^{(0)} = E_n^{(0)} w_n^{(0)}$

VALORE ESATTO. $w_n^{(0)} E_n^{(0)}$

$$\langle w_n^{(0)} | w_m^{(0)} \rangle = \int w_n^{(0)} w_m^{(0)} d\tau = \delta_{n,m} \quad , \quad \langle w_n^{(0)} | \hat{H}^{(0)} | w_m^{(0)} \rangle = E_n^{(0)} \delta_{n,m}$$

E_s.: BUCIA POT. INF.

$$w_n^{(0)}(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) \quad n = 1, 2, 3 \quad E_n^{(0)} = \frac{\pi^2 \hbar^2 n^2}{2m_e a^2}$$



AGGIUNGO V_0

$$\hat{V}(x) = \begin{cases} V_0 & 0 \leq x \leq a/2 \\ 0 & \text{altrove} \end{cases}$$

PERTURBAZIONE $\hat{H}' = \hat{V}(x)$

SISTEMA PERTURB.: $\hat{H} = \hat{H}^{(0)} + \lambda \hat{H}'$

$$\hookrightarrow \hat{H} w_n = E_n w_n \Rightarrow \text{NON SI POSSO CASCARE}$$

$\lambda \rightarrow$ Parametro piccolo $\lambda \ll 1 \Rightarrow \lambda \hat{H}' \ll \hat{H}^{(0)} \Rightarrow \hat{V}_0$ piccolo rispetto al sistema

$$\Rightarrow \begin{cases} w_n = w_n^{(0)} + \lambda w_n^{(1)} + \lambda^2 w_n^{(2)} + \dots \\ E_n = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \dots \end{cases}$$

$\lambda^k E_n^{(k)} \rightarrow$ ORDINE della CORREZIONE
Pot. di λ

$$\hat{H} w_n = E_n w_n \Rightarrow$$

$$[\hat{H}^{(0)} + \lambda \hat{H}'] [w_n^{(0)} + \lambda w_n^{(1)} + \lambda^2 w_n^{(2)} + \dots] = [E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \dots] [w_n^{(0)} + \lambda w_n^{(1)} + \lambda^2 w_n^{(2)} + \dots]$$

$\rightarrow$ ESPANITO SOLO TERMINI ALLA II^a POT. in λ

$$\hat{H}^{(0)} w_n^{(0)} + \lambda \hat{H}^{(0)} w_n^{(1)} + \lambda^2 \hat{H}^{(0)} w_n^{(2)} + \lambda \hat{H}' w_n^{(0)} + \lambda^2 \hat{H}' w_n^{(1)} + \dots =$$

$$= E_n^{(0)} w_n^{(0)} + \lambda E_n^{(0)} w_n^{(1)} + \lambda^2 E_n^{(0)} w_n^{(2)} + \lambda E_n^{(1)} w_n^{(0)} + \lambda^2 E_n^{(1)} w_n^{(1)} + \lambda^2 E_n^{(2)} w_n^{(0)} + \dots$$

$\Rightarrow$ RACCOLGO i term. sulle base delle pot. di λ

$$\begin{aligned}
 &= \underbrace{\hat{H}^{(0)} w_n^{(0)}}_{\lambda^0} + \underbrace{\lambda [\hat{H}^{(0)} w_n^{(1)} + \hat{H}' w_n^{(0)}]}_{1^{\text{ste pot}} \lambda} + \underbrace{\lambda^2 [\hat{H}^{(0)} w_n^{(2)} + \hat{H}' w_n^{(1)}]}_{2^{\text{te pot}} \lambda} + \dots = \\
 &= \underbrace{E_n^{(0)} w_n^{(0)}}_{\lambda^0} + \underbrace{\lambda [E_n^{(0)} w_n^{(1)} + E_n^{(1)} w_n^{(0)}]}_{1^{\text{ste pot}} \lambda} + \underbrace{\lambda^2 [E_n^{(0)} w_n^{(2)} + E_n^{(1)} w_n^{(1)} + E_n^{(2)} w_n^{(0)}]}_{2^{\text{te pot}} \lambda} + \dots
 \end{aligned}$$

$$\boxed{\lambda^0} \quad \hat{H}^{(0)} w_n^{(0)} = E_n^{(0)} w_n^{(0)}$$

$$\boxed{\lambda^1} \quad \hat{H}^{(0)} w_n^{(1)} + \hat{H}' w_n^{(0)} = E_n^{(0)} w_n^{(1)} + E_n^{(1)} w_n^{(0)}$$

$$\boxed{\lambda^2} \quad \hat{H}^{(0)} w_n^{(2)} + \hat{H}' w_n^{(1)} = E_n^{(0)} w_n^{(2)} + E_n^{(1)} w_n^{(1)} + E_n^{(2)} w_n^{(0)}$$

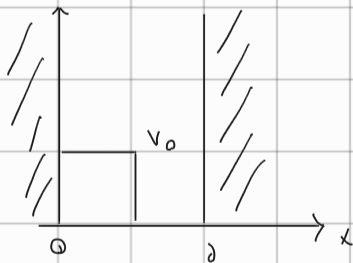
$$\boxed{\lambda^1} \quad \hat{H}^{(0)} w_n^{(1)} + \hat{H}' w_n^{(0)} = E_n^{(0)} w_n^{(1)} + E_n^{(1)} w_n^{(0)} \quad \text{Multipl. per } w_n^{(0)*} \text{ e integ.}$$

$$\begin{aligned}
 \Rightarrow \langle w_n^{(0)} | \hat{H}^{(0)} | w_n^{(1)} \rangle + \langle w_n^{(0)} | \hat{H}' | w_n^{(0)} \rangle &= E_n^{(0)} \langle w_n^{(0)} | w_n^{(1)} \rangle + E_n^{(1)} \underbrace{\langle w_n^{(0)} | w_n^{(0)} \rangle}_1 \\
 \hookrightarrow \langle \hat{H}^{(0)} w_n^{(0)} | w_n^{(1)} \rangle &= E_n^{(0)} \langle w_n^{(0)} | w_n^{(1)} \rangle
 \end{aligned}$$

$$\Rightarrow \cancel{E_n^{(0)} \langle w_n^{(0)} | w_n^{(1)} \rangle} + \langle w_n^{(0)} | \hat{H}' | w_n^{(0)} \rangle = \cancel{E_n^{(0)} \langle w_n^{(0)} | w_n^{(1)} \rangle} + E_n^{(1)}$$

$$\Rightarrow \boxed{E_n^{(1)} = \langle w_n^{(0)} | \hat{H}' | w_n^{(0)} \rangle} = \int_{\mathbb{R}} w_n^{(0)*} \hat{H}' w_n^{(0)} dx$$

Ex.:



$$\hat{H}^{(0)} = \frac{\hat{p}^2}{2m}$$

$$w_n^{(0)} = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) \quad n = 1, 2, 3, \dots$$

$$E_n^{(0)} = \frac{\pi^2 \hbar^2 n^2}{2m_e a^2}$$

$$\hat{H}' = V = \begin{cases} V_0 & 0 \leq x \leq a/2 \\ 0 & \text{ALTRONDE} \end{cases}$$

$$\Rightarrow E_n^{(1)} = \int w_n^{(0)*} \hat{H}' w_n^{(0)} dx = \int_0^{a/2} w_n^{(0)*} V_0 w_n^{(0)} dx$$

$$= V_0 \int_0^{a/2} |w_n^{(0)}|^2 dx = V_0 \int_0^{a/2} \frac{2}{a} \sin^2\left(\frac{n\pi x}{a}\right) dx = \frac{V_0}{2}$$

$$\Rightarrow E_1^{(1)} = V_0 \int_0^{a/2} \frac{2}{a} \sin^2\left(\frac{\pi x}{a}\right) dx = \frac{V_0}{2}$$

$$E_n = E_n^{(0)} + \lambda \frac{V_0}{2} \Rightarrow E_n = \frac{\hbar^2 n^2 \pi^2}{2 m_e a^2} + \frac{V_0}{2}$$

poiché $\lambda = 1$

AUTOVALORE (IMPERTURBATO) CORRETTO APPROSSIMATO

① DOMINIO di $\hat{H}'$ SI DEVE SOVRAPPORRE ad $\psi_n^{(0)}$

② PARITÀ $\psi_n^{(0)}$ E $\hat{H}'$ HANNO PARITÀ DIFFERENTE $\Rightarrow \int dx = 0$

Autofunzioni:

λ'

$$\hat{H}^{(0)} \psi_n^{(1)} + \hat{H}' \psi_n^{(0)} = E_n^{(0)} \psi_n^{(1)} + E_n^{(1)} \psi_n^{(0)}$$

$$\underbrace{[\hat{H}^{(0)} - E_n^{(0)}]}_{\text{OPERAT.}} \psi_n^{(1)} = - \underbrace{[\hat{H}' - E_n^{(1)}]}_{\text{TERMINE NOTO}} \psi_n^{(0)}$$

$\psi_n^{(0)} \Rightarrow$ BASE COMPLETA $\Rightarrow \psi_n^{(1)} = \sum_m c_m \psi_m^{(0)}$ (comb. lineare di tutte le autof. impert.)
(lo posso fare perché $\psi_n^{(1)}$ vive nello stesso spazio di Hilbert)

$$\psi_n^{(0)} \in \text{Ker} \{ \hat{H}^{(0)} - E_n^{(0)} \} \Rightarrow [\hat{H}^{(0)} - E_n^{(0)}] \psi_n^{(0)} = 0$$

Visto che $\psi_n^{(0)} \in \text{Ker}$ posso sottrarlo a $\psi_n^{(1)}$ e il risult. non cambia

$$\Rightarrow \boxed{\psi_n^{(1)} = \sum_{m \neq n} c_m \psi_m^{(0)}}$$

$$E_n^{(1)} = \langle w_n^{(0)} | \hat{H}' | w_n^{(0)} \rangle \quad [\hat{H}^{(0)} - E_n^{(0)}] w_n^{(1)} = - [\hat{H}' - E_n^{(1)}] w_n^{(0)} \Rightarrow w_n^{(1)} = \sum_{m \neq n} c_m w_m^{(0)}$$

$$[\hat{H}^{(0)} - E_n^{(0)}] \sum_{m \neq n} c_m w_m^{(0)} = - [\hat{H}' - E_n^{(1)}] w_n^{(0)} \Rightarrow \sum_{m \neq n} c_m \hat{H}^{(0)} w_m^{(0)} =$$

$$- \sum_{m \neq n} E_m^{(0)} c_m w_m^{(0)} = - [\hat{H}' - E_n^{(1)}] w_n^{(0)}$$

$$\Rightarrow \sum_{m \neq n} E_m^{(0)} w_m^{(0)} - \sum_{m \neq n} E_m^{(0)} c_m w_m^{(0)} = - [\hat{H}' - E_n^{(1)}] w_n^{(0)}$$

$$\Rightarrow \sum_{m \neq n} (E_m^{(0)} - E_n^{(0)}) c_m w_m^{(0)} = - [\hat{H}' - E_n^{(1)}] w_n^{(0)} \quad \text{Moltip. tutto per } w_\ell^{(0)*}$$

$$\Rightarrow \sum_{m \neq n} (E_m^{(0)} - E_n^{(0)}) c_m \langle w_\ell^{(0)} | w_m^{(0)} \rangle = - \langle w_\ell^{(0)} | \hat{H}' | w_n^{(0)} \rangle + E_n^{(1)} \langle w_\ell^{(0)} | w_n^{(0)} \rangle$$

$$\bullet \boxed{\ell = n} \Rightarrow \langle w_\ell^{(0)} | w_n^{(0)} \rangle = 0 \Rightarrow 0 = - \langle w_n^{(0)} | \hat{H}' | w_n^{(0)} \rangle + E_n^{(1)} \underbrace{\langle w_n^{(0)} | w_n^{(0)} \rangle}_1$$

$$\Rightarrow E_n^{(1)} = \langle w_n^{(0)} | \hat{H}' | w_n^{(0)} \rangle$$

$$\bullet \boxed{\ell \neq n} \Rightarrow \langle w_\ell^{(0)} | w_n^{(0)} \rangle \text{ solo i sott. con } \ell = n$$

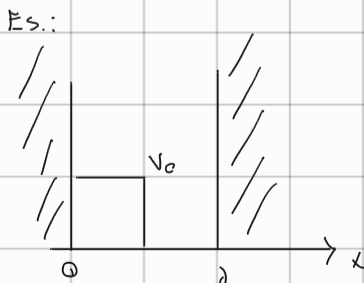
$$\Rightarrow (E_\ell^{(0)} - E_n^{(0)}) c_\ell = - \langle w_\ell^{(0)} | \hat{H}' | w_n^{(0)} \rangle + E_n^{(1)} \cancel{\langle w_\ell^{(0)} | w_n^{(0)} \rangle}$$

$$\Rightarrow c_\ell = \frac{\langle w_\ell^{(0)} | \hat{H}' | w_n^{(0)} \rangle}{E_n^{(0)} - E_\ell^{(0)}}$$

$$w_n^{(1)} = \sum_{m \neq n} \frac{\langle w_m^{(0)} | \hat{H}' | w_n^{(0)} \rangle}{E_n^{(0)} - E_m^{(0)}} w_m^{(0)}$$

(h) $\rightarrow$ AUTOSTATO CHE STO CORREGGENDO / AUTOS. DI CUI VOGLIO TROV. LA CORR.

(m) $\rightarrow$ AUTOST. CON CUI CORREGGO n



$$w_n^{(0)} = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)$$

$$E_n^{(0)} = \frac{\hbar^2 n^2 \pi^2}{2 m_e a^2}$$

$$w_1^{(1)} = \sum_{n=1} \frac{\langle w_2^{(0)} | \hat{H}' | w_1^{(0)} \rangle}{E_1^{(0)} - E_2^{(0)}} w_2^{(0)} + \frac{\langle w_3^{(0)} | \hat{H}' | w_1^{(0)} \rangle}{E_1^{(0)} - E_3^{(0)}} w_3^{(0)} +$$

(bracket under the first term)

$$+ \frac{\langle \psi_n^{(0)} | \hat{H}' | \psi_1^{(0)} \rangle}{E_1^{(0)} - E_n^{(0)}} \psi_n^{(0)} + \dots$$

$$c_2 = \frac{\langle \psi_2^{(0)} | \hat{H}' | \psi_1^{(0)} \rangle}{E_1^{(0)} - E_2^{(0)}} = \frac{\int_0^{a/2} \frac{2}{a} \sin\left(\frac{2\pi x}{a}\right) V_0 \sin\left(\frac{\pi x}{a}\right) dx}{\frac{\hbar^2 \pi^2}{2m_e a^2} - \frac{\hbar^2 \pi^2 4}{2m_e a^2}} = \frac{(4V_0/3\pi)}{-\frac{3\hbar^2 \pi^2}{2m_e a^2}}$$

$$c_3 = \frac{\langle \psi_3^{(0)} | \hat{H}' | \psi_1^{(0)} \rangle}{E_1^{(0)} - E_3^{(0)}} = \frac{\int_0^{a/2} \frac{2}{a} \sin\left(\frac{3\pi x}{a}\right) V_0 \sin\left(\frac{\pi x}{a}\right) dx}{\frac{\hbar^2 \pi^2}{2m_e a^2} - \frac{\hbar^2 \pi^2 9}{2m_e a^2}} = 0$$

$$c_4 = \frac{\langle \psi_4^{(0)} | \hat{H}' | \psi_1^{(0)} \rangle}{E_1^{(0)} - E_4^{(0)}} = \frac{\int_0^{a/2} \frac{2}{a} \sin\left(\frac{4\pi x}{a}\right) V_0 \sin\left(\frac{\pi x}{a}\right) dx}{\frac{\hbar^2 \pi^2}{2m_e a^2} - \frac{\hbar^2 \pi^2 16}{2m_e a^2}} = \frac{(-8V_0/15\pi)}{-\frac{15\hbar^2 \pi^2}{2m_e a^2}}$$

$$\psi_1 = \underbrace{\sqrt{\frac{2}{a}} \sin\left(\frac{\pi x}{a}\right)}_{\psi_1^{(0)}} - \underbrace{\left(\frac{8V_0 m_e a^2}{9\hbar^2 \pi^2}\right) \sqrt{\frac{2}{a}} \sin\left(\frac{2\pi x}{a}\right)}_{c_2 \psi_2^{(0)}} + \underbrace{\left(\frac{16V_0 m_e a^2}{225\hbar^2 \pi^2}\right) \sqrt{\frac{2}{a}} \sin\left(\frac{4\pi x}{a}\right)}_{c_4 \psi_4^{(0)}} + \dots$$

CORREZIONE al 2° ordine Ener.:

$$\boxed{\lambda^2} \quad \hat{H}^{(0)} \psi_n^{(2)} + \hat{H}^{(1)} \psi_n^{(1)} = E_n^{(0)} \psi_n^{(2)} + E_n^{(1)} \psi_n^{(1)} + E_n^{(2)} \psi_n^{(0)} \quad \text{Moltip. per } \psi_n^{(0)*}$$

$$\underbrace{\langle \psi_n^{(0)} | \hat{H}^{(0)} | \psi_n^{(2)} \rangle + \langle \psi_n^{(0)} | \hat{H}^{(1)} | \psi_n^{(1)} \rangle}_{\downarrow} = E_n^{(0)} \underbrace{\langle \psi_n^{(0)} | \psi_n^{(2)} \rangle}_{\downarrow} + E_n^{(1)} \langle \psi_n^{(0)} | \psi_n^{(1)} \rangle + E_n^{(2)} \langle \psi_n^{(0)} | \psi_n^{(0)} \rangle$$

$$\langle \psi_n^{(0)} | \psi_n^{(0)} \rangle = 1 \quad \langle \psi_n^{(0)} | \psi_n^{(1)} \rangle = 0 \quad \langle \psi_n^{(0)} | \psi_n^{(2)} \rangle = 0$$

$$\Rightarrow E_n^{(2)} = \langle \psi_n^{(0)} | \hat{H}^{(1)} | \psi_n^{(1)} \rangle - E_n^{(1)} \langle \psi_n^{(0)} | \psi_n^{(1)} \rangle$$

$$\Rightarrow \langle \psi_n^{(0)} | \psi_n^{(1)} \rangle = \langle \psi_n^{(0)} | \sum_{m \neq n} c_m \psi_m^{(0)} \rangle = 0$$

$$\Rightarrow \boxed{E_n^{(2)} = \langle \psi_n^{(0)} | \hat{H}^{(1)} | \psi_n^{(1)} \rangle}$$

$$E_n^{(2)} = \langle w_n^{(0)} | \hat{H}' \sum_{m \neq n} \frac{c_m w_m^{(0)}}{E_n^{(0)} - E_m^{(0)}} \rangle = \sum_{m \neq n} c_m \langle w_n^{(0)} | \hat{H}' | w_m^{(0)} \rangle =$$

$$= \sum_{m \neq n} \frac{\langle w_m^{(0)} | \hat{H}' | w_n^{(0)} \rangle \langle w_n^{(0)} | \hat{H}' | w_m^{(0)} \rangle}{E_n^{(0)} - E_m^{(0)}} =$$

$$\Rightarrow E_n^{(2)} = \sum_{m \neq n} \frac{|\langle w_m^{(0)} | \hat{H}' | w_n^{(0)} \rangle|^2}{E_n^{(0)} - E_m^{(0)}}$$

$n \Rightarrow$ STATO CHE CORREGGO

$m \Rightarrow$ UNO DEGLI STATI CON CUI CORREGGO

n

1) Suppongo n stato fond.: $E_n^{(0)} - E_m^{(0)} < 0$

$$\Rightarrow E_{SF}^{(2)} < 0$$

2) $\langle w_m^{(0)} | \hat{H}' | w_n^{(0)} \rangle \rightarrow$ Tutti dello stesso ordine di grandezza

$\Rightarrow$ ciò che conta veramente è il denominatore, più lo stato con cui

voglio correggere è lontano da n più la correzione è piccola

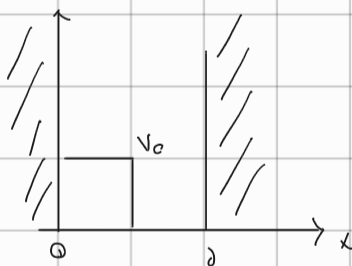
3) Se lo stato m sta sopra $n \Rightarrow E_n^{(0)} - E_m^{(0)} < 0 \Rightarrow$ lo stato m

spinge verso il basso n . Se $m < n \Rightarrow m$ spinge verso l'alto

$\Rightarrow$ TENDENZA dei LIV. ENERGETICI A RESPINGERSI

$\hookrightarrow$ No level-crossing theorem

Es.



$$w_n^{(0)} = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)$$

$$E_n^{(0)} = \frac{\hbar^2 n^2 \pi^2}{2 m_e a^2}$$

$$E_1^{(2)} = \frac{|\langle w_2^{(0)} | \hat{H}' | w_1^{(0)} \rangle|^2}{E_1^{(0)} - E_2^{(0)}} + \frac{|\langle w_3^{(0)} | \hat{H}' | w_1^{(0)} \rangle|^2}{E_1^{(0)} - E_3^{(0)}} + \frac{|\langle w_4^{(0)} | \hat{H}' | w_1^{(0)} \rangle|^2}{E_1^{(0)} - E_4^{(0)}} + \dots$$

$$|\langle w_2^{(0)} | \hat{H}' | w_1^{(0)} \rangle|^2 = \frac{4 V_0}{3\pi}, \quad |\langle w_3^{(0)} | \hat{H}' | w_1^{(0)} \rangle|^2 = 0, \quad |\langle w_4^{(0)} | \hat{H}' | w_1^{(0)} \rangle|^2 = -\frac{8 V_0}{15\pi}$$

$$\Rightarrow E_1^{(2)} = \frac{\left(\frac{4 V_0}{3\pi}\right)^2}{\frac{\hbar^2 \pi^2}{2 m_e a^2} - \frac{4 \hbar^2 \pi^2}{2 m_e a^2}} + 0 + \frac{\left(-\frac{8}{15\pi} V_0\right)^2}{\frac{\hbar^2 \pi^2}{2 m_e a^2} - \frac{16 \hbar^2 \pi^2}{2 m_e a^2}} + \dots$$

$$\Rightarrow E_1 = \frac{\hbar^2 \pi^2}{2m_0 d^2} + \frac{V_0}{2} - \frac{32 V_0^2 m_0 d^2}{27 \hbar^2 \pi^4} - \frac{128 V_0^3 m_0 d^2}{3375 \hbar^2 \pi^4}$$

$E_1^{(2)}$

SISTEMI DEGENERI:

$$\left. \begin{aligned} \hat{H}^{(0)} w_{na}^{(0)} &= E_n^{(0)} w_{na}^{(0)} \\ \hat{H}^{(0)} w_{nb}^{(0)} &= E_n^{(0)} w_{nb}^{(0)} \end{aligned} \right\} \Rightarrow \langle w_{na}^{(0)} | w_{nb}^{(0)} \rangle = 0$$

SISTEMI DEGENERI di Dimensione (2)

$$w_n^{(0)} = \alpha w_{na}^{(0)} + \beta w_{nb}^{(0)} \quad \alpha, \beta \in \mathbb{C}$$

$$\bullet E_n^{(1)} = \langle w_n^{(0)} | \hat{H}^{(1)} | w_n^{(0)} \rangle \Rightarrow \text{QUALE } w_n^{(0)} \text{ PRENDO?}$$

$$\bullet E_n^{(0)} - E_m^{(0)} = E_{na}^{(0)} - E_{nb}^{(0)} = 0 \Rightarrow w_n^{(1)} \rightarrow +\infty \Rightarrow E_n^{(1)} \rightarrow +\infty$$

$$\boxed{\hat{H} w_n = E_n w_n} \quad \hat{H} = \hat{H}^{(0)} + \lambda \hat{H}^{(1)} \Rightarrow \begin{cases} w_n = w_n^{(0)} + \lambda w_n^{(1)} + \lambda^2 w_n^{(2)} + \dots \\ E_n = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \dots \end{cases}$$

1° pot. in λ

$$[\hat{H}^{(0)} + \lambda \hat{H}^{(1)}] [w_n^{(0)} + \lambda w_n^{(1)}] = [E_n^{(0)} + \lambda E_n^{(1)}] [w_n^{(0)} + \lambda w_n^{(1)}]$$

$$\hat{H}^{(0)} w_n^{(0)} + \lambda \hat{H}^{(0)} w_n^{(1)} + \lambda \hat{H}^{(1)} w_n^{(0)} + \lambda^2 \hat{H}^{(1)} w_n^{(1)} = E_n^{(0)} w_n^{(0)} + \lambda E_n^{(0)} w_n^{(1)} + \lambda E_n^{(1)} w_n^{(0)} + \lambda^2 E_n^{(1)} w_n^{(1)}$$

$$\boxed{\lambda^1} \quad \hat{H}^{(0)} w_n^{(1)} + \hat{H}^{(1)} w_n^{(0)} = E_n^{(0)} w_n^{(1)} + E_n^{(1)} w_n^{(0)} \quad \text{MOLTIPL. PER } w_{na}^{(0)*} \text{ e integ.}$$

$$\Rightarrow \langle w_{na}^{(0)} | \hat{H}^{(0)} | w_n^{(1)} \rangle + \langle w_{na}^{(0)} | \hat{H}^{(1)} | w_n^{(0)} \rangle = E_n^{(0)} \langle w_{na}^{(0)} | w_n^{(1)} \rangle + E_n^{(1)} \langle w_{na}^{(0)} | w_n^{(0)} \rangle$$

$$\hookrightarrow \langle \hat{H}^{(0)} w_{na}^{(0)} | w_n^{(1)} \rangle = E_n^{(0)} \langle w_{na}^{(0)} | w_n^{(1)} \rangle$$

$$\Rightarrow E_n^{(0)} \langle w_{na}^{(0)} | w_n^{(1)} \rangle + \langle w_{na}^{(0)} | \hat{H}^{(1)} | w_n^{(0)} \rangle = E_n^{(0)} \langle w_{na}^{(0)} | w_n^{(1)} \rangle + E_n^{(1)} \langle w_{na}^{(0)} | w_n^{(0)} \rangle$$

$$\Rightarrow \langle w_{na}^{(0)} | \hat{H}^{(1)} | w_n^{(0)} \rangle = E_n^{(1)} \langle w_{na}^{(0)} | w_n^{(0)} \rangle, \quad w_n^{(0)} = \alpha w_{na}^{(0)} + \beta w_{nb}^{(0)}$$

$$\langle w_{na}^{(0)} | \hat{H}^{(1)} | \alpha w_{na}^{(0)} + \beta w_{nb}^{(0)} \rangle = E_n^{(1)} \langle w_{na}^{(0)} | \alpha w_{na}^{(0)} + \beta w_{nb}^{(0)} \rangle$$

$$\Rightarrow \alpha \langle w_{na}^{(0)} | \hat{H}^{(1)} | w_{na}^{(0)} \rangle + \beta \langle w_{na}^{(0)} | \hat{H}^{(1)} | w_{nb}^{(0)} \rangle = \alpha E_n^{(1)}$$

Posso fare la stessa cosa per $w_{nb}^{(0)}$ e ottengo:

$$\Rightarrow \alpha \langle w_{nb}^{(0)} | \hat{H}^{(1)} | w_{na}^{(0)} \rangle + \beta \langle w_{nb}^{(0)} | \hat{H}^{(1)} | w_{nb}^{(0)} \rangle = \beta E_n^{(1)}$$

$$\Rightarrow \begin{pmatrix} \langle w_{n_a}^{(0)} | \hat{H}' | w_{n_a}^{(0)} \rangle & \langle w_{n_a}^{(0)} | \hat{H}' | w_{n_b}^{(0)} \rangle \\ \langle w_{n_b}^{(0)} | \hat{H}' | w_{n_a}^{(0)} \rangle & \langle w_{n_b}^{(0)} | \hat{H}' | w_{n_b}^{(0)} \rangle \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = E_n^{(1)} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$\begin{pmatrix} \hat{H}'_{aa} & \hat{H}'_{ab} \\ \hat{H}'_{ba} & \hat{H}'_{bb} \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = E_n^{(1)} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

MATRICE dell'op $\hat{H}'$ scritta sulla base w_{n_a} e w_{n_b}

↳ valore della comb. lin. $\alpha w_{n_a}^{(0)} + \beta w_{n_b}^{(0)}$

$\Rightarrow$ Se diagonali, la mat. trova le cor. al 1° ordine dell' E_n .

Autov. di $\hat{H}'$ scritta nel sottosp. degen. $\Rightarrow E_n^{(1)}$

Es.: δ_1, δ_2 (i due subsp. che trovo)

$$\begin{array}{l} E_n^{(0)} \\ \swarrow \quad \searrow \\ E_1 = E_n^{(0)} + \delta_1 \\ E_1 = E_n^{(0)} + \delta_2 \end{array}$$

SE considero $\hat{H}'$ rimuovo la degenerazione

$$\Rightarrow \det(\delta I - \hat{H}') = 0$$

$$H'_{ab} = H'_{ba}^*$$

$$\det \begin{pmatrix} \delta_1 - H'_{aa} & -H'_{ab} \\ -H'_{ba} & \delta_1 - H'_{bb} \end{pmatrix} \Rightarrow \delta_1, \delta_2$$

$$\delta_1 \Rightarrow \begin{pmatrix} \delta_1 - H'_{aa} & -H'_{ab} \\ -H'_{ba} & \delta_1 - H'_{bb} \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = 0 \Rightarrow \alpha_1, \beta_1 \Rightarrow \boxed{w_{n_1}^{(0)} = \alpha_1 w_{n_a}^{(0)} + \beta_1 w_{n_b}^{(0)}}$$

$\rightarrow$ Faccio uguale per $w_{n_2}^{(0)}$

$$E_n = E_n^{(0)} + \delta_1 \Rightarrow w_{n_1}^{(0)} = \alpha_1 w_{n_a}^{(0)} + \beta_1 w_{n_b}^{(0)}$$

$$\begin{array}{l} E_n^{(0)} \\ \swarrow \quad \searrow \\ \end{array}$$

$$E_n = E_n^{(0)} + \delta_2 \Rightarrow w_{n_2}^{(0)} = \alpha_2 w_{n_a}^{(0)} + \beta_2 w_{n_b}^{(0)}$$

$$w_{n_1}^{(0)} = \alpha_1 w_{n_a}^{(0)} + \beta_1 w_{n_b}^{(0)} \Rightarrow \begin{pmatrix} \alpha_1 \\ \beta_1 \end{pmatrix} \hookrightarrow \delta_1$$

$$w_{n_2}^{(0)} = \alpha_2 w_{n_a}^{(0)} + \beta_2 w_{n_b}^{(0)} \Rightarrow \begin{pmatrix} \alpha_2 \\ \beta_2 \end{pmatrix} \hookrightarrow \delta_2$$

NON È UNA CORREZ. di $w_n^{(0)}$

$w_{n_1}^{(0)}$ e $w_{n_2}^{(0)}$ SONO AUTOSTATI di $\hat{H}^{(0)}$

Quindi in questo modo per gli stati degeneri ottengo una correzione 1° ord. sull'En. che mi fa capire se i livelli si dividono o no e capisco il modo in cui gli stati imperturbati si mescolano tra loro.

CONOSCO A PRIORI (α_1, β_1) (α_2, β_2)

$$\begin{pmatrix} \langle \psi_{n_1}^{(0)} | \hat{H}' | \psi_{n_1}^{(0)} \rangle & 0 \\ 0 & \langle \psi_{n_2}^{(0)} | \hat{H}' | \psi_{n_2}^{(0)} \rangle \end{pmatrix} \Rightarrow \text{la matrice è diagonale}$$

$\Rightarrow \psi_{n_1}^{(0)} \text{ e } \psi_{n_2}^{(0)} \Rightarrow$ BUONI STATI (quelli che diagonalizzano la matrice)

$$\left. \begin{aligned} \delta_1 &= \langle \psi_{n_1}^{(0)} | \hat{H}' | \psi_{n_1}^{(0)} \rangle \\ \delta_2 &= \langle \psi_{n_2}^{(0)} | \hat{H}' | \psi_{n_2}^{(0)} \rangle \end{aligned} \right\} \text{TEORIA NON DEGENERE}$$

sono in grado di conoscere a priori i BUONI STATI!

TEOREMA:

cons. $\hat{O} = \hat{O}^\dagger \Rightarrow [\hat{O}, \hat{H}'] = 0$ se $\psi_{n_1}^{(0)}$ e $\psi_{n_2}^{(0)}$ sono AUTOSTATI di $\hat{O}$ con

AUTOSTATI DISTINTI: $\left. \begin{aligned} \hat{O} \psi_{n_1}^{(0)} &= \xi \psi_{n_1}^{(0)} \\ \hat{O} \psi_{n_2}^{(0)} &= \nu \psi_{n_2}^{(0)} \end{aligned} \right\} \Rightarrow \boxed{H'_{12} = 0} \text{ la matrice è diag.}$

Dim.: $[\hat{O}, \hat{H}] = 0$

$$\begin{aligned} \langle \psi_{n_1}^{(0)} | [\hat{O}, \hat{H}] | \psi_{n_2}^{(0)} \rangle &= 0 = \langle \psi_{n_1}^{(0)} | \hat{O} \hat{H}' | \psi_{n_2}^{(0)} \rangle - \langle \psi_{n_1}^{(0)} | \hat{H}' \hat{O} | \psi_{n_2}^{(0)} \rangle = \\ &= \langle \hat{O} \psi_{n_1}^{(0)} | \hat{H}' | \psi_{n_2}^{(0)} \rangle - \langle \psi_{n_1}^{(0)} | \hat{H}' | \hat{O} \psi_{n_2}^{(0)} \rangle = \xi \langle \psi_{n_1}^{(0)} | \hat{H}' | \psi_{n_2}^{(0)} \rangle - \nu \langle \psi_{n_1}^{(0)} | \hat{H}' | \psi_{n_2}^{(0)} \rangle = \\ &= (\xi - \nu) \underbrace{\langle \psi_{n_1}^{(0)} | \hat{H}' | \psi_{n_2}^{(0)} \rangle}_1 = 0 \\ &\quad \quad \quad \hookrightarrow = 0 \end{aligned}$$

■

δ^1

$$\begin{pmatrix} \langle \psi_{n_1}^{(0)} | \hat{H}' | \psi_{n_1}^{(0)} \rangle & 0 \\ 0 & \underbrace{\langle \psi_{n_2}^{(0)} | \hat{H}' | \psi_{n_2}^{(0)} \rangle}_{\delta^2} \end{pmatrix}$$

DEGENERAZ. di ORDINE SUPERIORE:

$$\hat{H}^{(1)}_{nn_0} = E_n^{(0)} w_{nn_0}^{(1)}$$

$$\hat{H}^{(1)}_{nn_1} = E_n^{(0)} w_{nn_1}^{(1)}$$

$\vdots$

$$\hat{H}^{(1)}_{nn_k} = E_n^{(0)} w_{nn_k}^{(1)}$$

① IDENTICO SOTTO SP. DEGENERE

$$w_n^{(1)} = a w_{n_0}^{(1)} + \dots + k w_{n_k}^{(1)}$$

$$\textcircled{2} \begin{pmatrix} H'_{n_0 n_0} & \dots & H'_{n_0 k} \\ \vdots & & \vdots \\ H'_{k n_0} & \dots & H'_{k k} \end{pmatrix} \begin{pmatrix} a \\ b \\ \vdots \\ k \end{pmatrix} = E_n^{(1)} \begin{pmatrix} a \\ b \\ \vdots \\ k \end{pmatrix}$$

② DIAGONALIZZO